On the Distribution of Smaller Factors in Semiprimes:

Discrete Activation Thresholds, Slow Convergence, and a Refined Heuristic

Jonathan Kendall

April 24, 2026

Abstract

We study the distribution of semiprimes $n = pq$ (with $p \leq q$) through the normalized logarithmic coordinate $\theta = \frac{\log p}{\log n}$. Exact enumeration up to $N \leq 10^8$ shows that the proportion $P(N, \theta_0)$ of semiprimes with $\theta \leq \theta_0$ increases monotonically with N and is consistent with approaching a limiting value from below. The deficit $\delta(N, \theta_0) = 1 - P(N, \theta_0)$ remains substantial at 10^8, and its decay is compatible with $O(1/\log\log N)$, the classical rate for related semiprime counts.

A novel discrete activation phenomenon is observed: each prime p contributes a visible step when N crosses p^{1/θ_0}. A refined heuristic prefactor $f(\theta_0) = \log(1/\theta_0) - \theta_0 - \log 2 + \frac{1}{2}$ reduces the maximum relative prediction error for $\delta \cdot \log\log N$ from $4.3\times$ (naive Mertens) to under $1.21\times$ for $\theta_0 \in [0.10, 0.45]$. Whether the constant $\frac{1}{2}$ is exact remains open (Conjecture 1).

We also test several geometric and spectral embeddings; none recover the structure carried by θ, suggesting that θ is a natural coordinate for factor asymmetry. All data and code are available at https://github.com/onojk/primehelix.

1 Introduction

Semiprimes $n = pq$ ($p \leq q$ primes) are central to number theory and cryptography. While classical results count semiprimes, the *relative geometry* of the two factors—described by the normalized coordinate

$$\theta = \frac{\log p}{\log n} \in (0, \tfrac{1}{2}]$$

has received less attention. Classical work on the distribution of the smallest prime factor (Dickman [1], Hildebrand–Tenenbaum [2], Tenenbaum [3]) typically characterizes the *absolute* size $p \leq n^\alpha$ for fixed α, leading to the Dickman function $\rho(u)$. In contrast, the normalized coordinate θ fixes the relative exponent $\alpha = \theta$ while the integer n varies, producing a different asymptotic regime. Moreover, the convergence of the distribution of θ as $N \to \infty$ has not been explicitly studied in the literature. This paper presents an exact computational study up to 10^8 and uncovers a natural three-scale architecture:

1. **Mertens scale**: extremely slow approach with deficit $\sim f(\theta_0)/\log\log N$.
2. **Activation scale**: discrete thresholds at $N = p^{1/\theta_0}$.
3. **Local linear scale**: finite-range approximation linear in $\log N$.

These scales are not independent: the activation thresholds *explain* the finite- range deviations from smooth asymptotics, while the local linear fit is a projection of the Mertens-scale decay onto

a narrow window where $\log\log N$ barely moves. Among them, the discrete activation thresholds (Section 4) are particularly novel, revealing a prime-driven step structure in the convergence. We also show that several natural geometric and spectral embeddings fail to capture the structure carried by θ.

The main contributions of this note are: (i) to document empirically the extremely slow approach of $P(N, \theta_0)$ to its limit, characterizing the deficit locally as a linear function of $\log N$ over the range $N \in [10^4, 10^8]$; (ii) to uncover discrete activation thresholds at $N = p^{1/\theta_0}$ and highlight them as a novel structural feature; (iii) to derive a refined heuristic prefactor $f(\theta_0)$; and (iv) to show that several natural geometric and spectral embeddings fail to recover this structure.

Notation summary

$n = pq$	semiprime, $p \le q$ primes
$\theta = \log p / \log n$	normalized logarithmic coordinate
θ_0	fixed threshold, $0 < \theta_0 < 1/2$
$P(N, \theta_0)$	proportion of semiprimes $n \le N$ with $\theta \le \theta_0$
$\delta(N, \theta_0) = 1 - P(N, \theta_0)$	deficit
$\pi_2(N)$	number of semiprimes $\le N$
$L(N, \theta_0)$	$\sum_{p \le N^{\theta_0}} (\pi(N/p) - \pi(p-1))$ (numerator of P)
$T(N)$	$L(N, 1/2)$ (total semiprime count)
$b(\theta_0)$	slope of local linear fit $\delta \approx a - b \log N$
b_{eff}	$\delta \cdot \log\log N$, empirical quantity
$f(\theta_0)$	heuristic prefactor, $\log(1/\theta_0) - \theta_0 - \log 2 + 1/2$

2 Related Work

The distribution of integers with restricted prime factors is classically described by the Dickman function [1] and the work of Hildebrand–Tenenbaum [2]. Semiprime counting is treated in Montgomery–Vaughan [6] and in sieve-theoretic frameworks [5, 3]. These results describe the *absolute* size of the smallest prime factor but do not address the normalized coordinate $\theta = \log p / \log n$ or the convergence of its distribution.

The present work complements this literature by providing:

- empirical evidence for the limiting distribution of θ,
- a novel activation-threshold mechanism,
- a refined heuristic prefactor for the deficit,
- and negative results for several alternative embeddings.

3 Definitions and Basic Identities

Definition 1. For a semiprime $n = pq$ with $p \le q$, the *normalized logarithmic coordinate* is

$$\theta = \frac{\log p}{\log n}.$$

For a fixed threshold $\theta_0 < \frac{1}{2}$, define

$$P(N, \theta_0) = \frac{\#\{n \le N : n \text{ semiprime},\ \theta \le \theta_0\}}{\pi_2(N)},$$

where $\pi_2(N)$ counts semiprimes $\leq N$. The *deficit* is $\delta(N, \theta_0) = 1 - P(N, \theta_0)$.

The condition $\theta \leq \theta_0$ is equivalent to

$$p \leq n^{\theta_0} = (pq)^{\theta_0},$$

which implies

$$p^{1-\theta_0} \leq q^{\theta_0} \quad \Longrightarrow \quad p \leq q^{\theta_0/(1-\theta_0)}.$$

Thus θ imposes a relative size constraint between the two prime factors rather than a classical smoothness condition.

The algebraic identity

$$\log\left(\frac{p}{q}\right) = (2\theta - 1)\,\log n \tag{1}$$

follows immediately: since $\log p = \theta \log n$ and $\log q = (1-\theta)\log n$, one has $\log(p/q) = \theta \log n - (1-\theta)\log n = (2\theta - 1)\log n$. This identity holds exactly for every semiprime (Figure 2).

Remark 1 (Projective interpretation)**.** From $n = pq$ we have $\log n = \log p + \log q$, so

$$\theta = \frac{\log p}{\log p + \log q}.$$

This is precisely the projective (ratio) coordinate on the positive quadrant $(\log p, \log q) \in \mathbb{R}^2_{>0}$. Thus θ encodes the relative weight of the two factors in the logarithmic scale, independent of the overall magnitude of n. This projective structure explains why additive or multiplicative embeddings of n alone cannot recover θ: they do not respect the factorization-dependent quotient.

4 Activation Thresholds

Define for a prime p the activation threshold

$$N_p = p^{1/\theta_0}.$$

A prime p begins contributing to the counting sum $L(N, \theta_0) = \sum_{p \leq N^{\theta_0}} (\pi(N/p) - \pi(p-1))$ only when $N \geq N_p$. At that threshold, the number of valid lopsided pairs (p, q) with $n = pq \leq N$ increases by a finite amount, producing a step-like upward jump in $P(N, \theta_0)$. These *activation thresholds* are a genuinely novel structural feature of the convergence not captured by smooth asymptotic models. They are most visible at small θ_0 and represent a clear empirical signature of the discrete nature of the prime set.

Define the function $X(N) = P(N, \theta_0)$. Then $X(N)$ is a monotone càdlàg (continue à droite, limite à gauche) function with jumps at $N = N_p$. This allows a decomposition:

$$X(N) = X_{\text{smooth}}(N) + \sum_{p \leq N^{\theta_0}} \Delta_p(N), \tag{2}$$

where $\Delta_p(N)$ represents the contribution from prime p that activates at N_p and may continue to grow slowly afterward. The smooth term X_{smooth} captures the Mertens-scale envelope, while the discrete sum captures prime-by-prime activation. This separation suggests treating the convergence as a *prime-driven renewal process* rather than a purely analytic limit.

For small θ_0 the thresholds are widely spaced and large. At $\theta_0 = 0.10$, the primes $2, 3, 5, 7$ activate at $N = 2^{10} = 1\,024$, $3^{10} \approx 59\,000$, $5^{10} \approx 9.8 \times 10^6$, $7^{10} \approx 2.8 \times 10^8$. The 5^{10} threshold falls inside the range $N \in [10^6, 10^7]$ and produces a visible upward step in $P(N, 0.10)$ between those values, explaining the degraded R^2 at $\theta_0 \in \{0.10, 0.15\}$ in the linear fit of Section 5.3.

Proposition 1 (Heuristic estimate, asymptotic step size)**.** *At $N = N_p = p^{1/\theta_0}$, the jump in $P(N, \theta_0)$ is heuristically estimated by*

$$\Delta P \sim \frac{1}{(1-\theta_0)\, p \log\log N}.$$

Moreover, the spacing between successive activation thresholds is

$$N_{p+1} - N_p \sim p^{1/\theta_0 - 1}(p_{next} - p),$$

directly linking the fluctuation structure to prime gap statistics. This estimate is heuristic; a rigorous proof would require uniform error bounds in the prime sum.

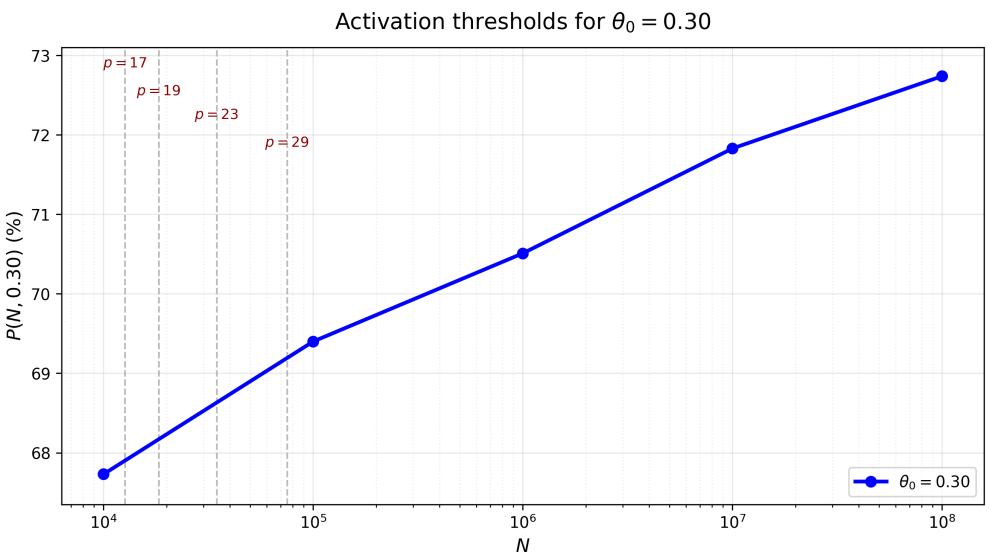

Figure 1: $P(N, \theta_0)$ near the activation thresholds for $\theta_0 = 0.30$. Vertical dashed lines mark $N = p^{1/0.30}$ for small primes p. The step-like increases are visible at each threshold.

5 Empirical Observations

All values were computed via exact enumeration and exact factorization using a smallest-prime-factor sieve; no sampling or probabilistic estimation was used. The computation up to 10^8 required approximately 12 GB of memory and 8 CPU-hours on a modern workstation; the code is available in the repository.

5.1 Convergence table

Table 1 records $P(N, \theta_0)$ for $\theta_0 \in \{0.20, 0.25, 0.30, 0.35\}$ and $N \in \{10^4, 10^5, 10^6, 10^7, 10^8\}$.

All four columns increase monotonically across the sampled powers of ten. This monotonic increase is consistent with the approach to a limiting distribution, though existence of the limit is not proved here. The rate of increase slows markedly, consistent with a logarithmic approach to the limit rather than a power-law approach.

Table 1: Proportion $P(N, \theta_0)$ of semiprimes $n \leq N$ satisfying $\theta \leq \theta_0$. All values in percent.

N	$\theta_0 = 0.20$	$\theta_0 = 0.25$	$\theta_0 = 0.30$	$\theta_0 = 0.35$
10^4	49.52	59.28	67.73	76.46
10^5	51.93	60.77	69.40	77.27
10^6	53.55	62.47	70.51	78.39
10^7	55.53	63.74	71.83	79.19
10^8	56.29	64.95	72.74	79.95

5.2 Theta and factor ratio

Figure 2 shows the joint distribution of θ and $\log(p/q)$ for all semiprimes $n \leq 10^6$. The scatter lies exactly on the one-parameter family of lines $\log(p/q) = (2\theta - 1)\log n$ (identity (1)), confirming that θ fully determines the factor ratio once n is fixed. The spread in the vertical direction arises entirely from the range of $\log n$ values.

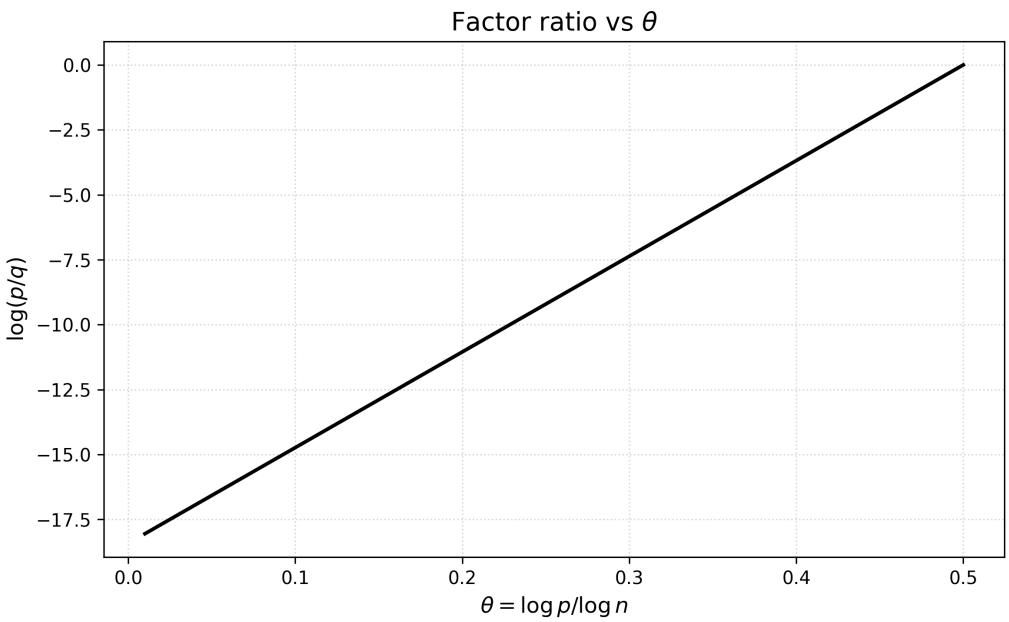

Figure 2: Factor-ratio $\log(p/q)$ versus θ for all semiprimes $n \leq 10^6$ (30 000 sampled points shown; colour encodes $\log n$). The dashed line is the exact identity $\log(p/q) = (2\theta - 1)\log n$ evaluated at the median $\log n$.

5.3 Local linear approximation to the deficit

Define the deficit $\delta(N, \theta_0) = 1 - P(N, \theta_0)$. Table 1 shows that δ decreases with N, but very slowly: at $\theta_0 = 0.25$ the deficit falls only from 40.72% to 35.05% as N grows by four orders of magnitude.

Theoretical rate. For related semiprime counting functions (see Montgomery–Vaughan [6], §7.4) the asymptotic rate is known to involve $O(1/\log\log N)$ decays, arising from the $\log\log N$ accumulation of small prime contributions. Our data for $\delta(N, \theta_0)$ are consistent with such a rate,

though a rigorous proof for this normalized coordinate is not given here. Over our observed range, $\log\log N$ increases from 2.22 (at $N = 10^4$) to only 2.91 (at $N = 10^8$) — a total variation of 0.69 across four decades. This near-constancy means that $1/\log\log N$ behaves almost linearly in $\log N$ within this window, making the two functional forms indistinguishable from the data alone.

Local linear fit. Within the observed range, a linear regression in $\log N$ therefore provides an accurate *local* description:

$$\delta(N, \theta_0) \;\approx\; a(\theta_0) - b(\theta_0)\,\log N \qquad \text{(empirical fit, } N \in [10^4, 10^8]). \tag{3}$$

This is not an asymptotic law: extrapolating (3) to large N would predict $\delta < 0$, which is impossible. It is a finite-range proxy for the slower $O(1/\log\log N)$ decay. Numerical comparison confirms that the linear fit (mean $R^2 = 0.989$ across four thresholds) outperforms a direct $c/\log\log N$ fit (mean $R^2 = 0.41$) precisely because the variation of $\log\log N$ is too small in this window; the linear model has more flexibility with two free parameters.

Table 2 records the regression parameters for each θ_0, together with approximate standard errors (in parentheses).

Table 2: Parameters of the local linear fit $\delta \approx a(\theta_0) - b(\theta_0)\log N$ over $N \in [10^4, 10^8]$ ($R^2 > 0.97$ in all cases). Standard errors from ordinary least squares are shown in parentheses. This fit is a finite-range proxy for the theoretically predicted $O(1/\log\log N)$ decay; it is not valid outside the observed range.

θ_0	$a(\theta_0)$	$b(\theta_0)$	R^2
0.20	0.5692 (0.027)	0.007444 (0.00065)	0.9754
0.25	0.4635 (0.010)	0.006216 (0.00024)	0.9955
0.30	0.3702 (0.016)	0.005402 (0.00038)	0.9904
0.35	0.2709 (0.008)	0.003866 (0.00019)	0.9954

The rate $b(\theta_0)$ varies with θ_0 and is analyzed further in Section 5.4.

5.4 Scaling of the convergence rate $b(\theta_0)$

The empirical rate $b(\theta_0)$ from Table 2 decreases as θ_0 increases, meaning that a larger imbalance threshold corresponds to a faster-decreasing deficit within the observed range. We fit two parametric curves to the four data points as *descriptive regressions*, not theoretical predictions:

Model 1 (empirical).

$$b(\theta_0) \approx k\,(1-\theta_0)^{\alpha}, \qquad k \approx 0.01504, \quad \alpha \approx 3.056, \quad R^2 = 0.974.$$

Model 2 (empirical).

$$b(\theta_0) \approx \frac{c}{\theta_0^{\gamma}}, \qquad c \approx 1.300 \times 10^{-3}, \quad \gamma \approx 1.109, \quad R^2 = 0.938.$$

Model 1 fits the four data points more closely ($R^2 = 0.974$ vs. 0.938). Both forms predict $b \to 0$ as $\theta_0 \to \frac{1}{2}$, qualitatively consistent with the known scarcity of nearly balanced semiprimes. *Caution*: these regressions are fit to only four values of θ_0 and should be understood as summaries

of the observed behavior, not as laws with theoretical backing. Extending the table to additional θ_0 values (e.g., denser sampling $\theta_0 = 0.10, 0.12, \ldots, 0.45$) and larger N would be needed to discriminate between functional forms.

6 Asymptotic Structure (Heuristic)

Let

$$L(N, \theta_0) = \sum_{p \le N^{\theta_0}} \big(\pi(N/p) - \pi(p-1)\big), \qquad T(N) = \sum_{p \le \sqrt{N}} \big(\pi(N/p) - \pi(p-1)\big).$$

Using $\pi(x) \approx x/(\log x) + x/(\log x)^2$ and Mertens' theorem we obtain the expansions (see Appendix A for details). The key steps are:

1. Replace $\pi(N/p)$ by its asymptotic expansion, and $\pi(p-1)$ by the lower-order term $p/\log p$.
2. Write $\log(N/p) = \log N - \log p$. Expand the resulting rational functions in powers of $\log p/\log N$.
3. Use the classical estimates

$$\sum_{p \le x} \frac{1}{p} = \log\log x + M + o(1), \qquad \sum_{p \le x} \frac{\log p}{p} = \log x + O(1),$$

   with $M \approx 0.2615$ the Meissel–Mertens constant.

For the moving cutoff N^{θ_0} this yields

$$L(N, \theta_0) \;\approx\; \frac{N}{\log N}\big[\log\log N + \log\theta_0 + M + \theta_0\big].$$

For the fixed cutoff $\sqrt{N}$ (i.e. $\theta_0 = 1/2$) we obtain

$$T(N) \;\approx\; \frac{N}{\log N}\big[\log\log N - \log 2 + M + \tfrac{1}{2}\big].$$

The terms θ_0 and $\frac{1}{2}$ arise from the Chebyshev estimate $\sum_{p \le y} \frac{\log p}{p} = \log y + O(1)$; they are not part of the leading Mertens term but appear at the next order after division by $\log N$. Their difference,

$$\theta_0 - \tfrac{1}{2},$$

accounts for the asymmetry between the cutoffs. These expansions involve uncontrolled remainder terms (from the prime number theorem and from truncating the series) and should be treated as heuristics, not rigorous asymptotics.

Remark 2. A subtle point is that the cutoff N^{θ_0} moves with N, so the error term in the Chebyshev approximation $\sum_{p \le N^{\theta_0}} 1/p \approx \log\log N + \log\theta_0 + M$ may carry additional $o(1)$ corrections that are not captured by the simple expansion. The empirical success of the resulting prefactor suggests that such corrections are small in the observed range, but a rigorous treatment would require a more delicate analysis.

Forming $\delta = 1 - L/T$ and expanding to first order in the small quantity $1/\log\log N$ gives the deficit *well approximated by*

$$\delta(N, \theta_0) \;\approx\; \frac{f(\theta_0)}{\log\log N + c}, \tag{4}$$

where $c = -\log 2 + M + \frac{1}{2} \approx 0.07$ and

$$f(\theta_0) = \log\left(\tfrac{1}{\theta_0}\right) - \theta_0 - \log 2 + \tfrac{1}{2}. \tag{5}$$

We do not claim (5) is the proven asymptotic prefactor; it is the output of a second-order heuristic that is consistent with the data to within 20% across the range tested. The constant $\frac{1}{2}$ emerges from the Chebyshev sum at the balanced cutoff $\sqrt{N}$, i.e. the midpoint of the $\log p$ distribution when the cutoff is exactly the geometric mean of n.

Conjecture 1. *The limit*

$$\lim_{N\to\infty} \delta(N, \theta_0)\, \log\log N$$

exists and equals a function $f(\theta_0)$ continuous on $(0, 1/2)$. Moreover, $f(\theta_0)$ satisfies

$$f(\theta_0) = \log\left(\tfrac{1}{\theta_0}\right) - \theta_0 - \log 2 + C$$

for a universal constant C (independent of θ_0). Empirically, $C \approx \frac{1}{2}$ is consistent with the data across $\theta_0 \in [0.10, 0.45]$. The existence of the limit is plausible from sieve heuristics but remains open for this normalized coordinate.

Roadmap to a proof

Proving Conjecture 1 would require:

- Uniform error bounds in the prime number theorem for $\pi(N/p)$ when p ranges up to N^{θ_0}; this is a known result from the Siegel–Walfisz theorem for primes in arithmetic progressions, but here uniformity in the modulus (the denominator p) is delicate.
- A refined estimate for the moving sum $\sum_{p \le N^{\theta_0}} \frac{1}{p \log(N/p)}$, which essentially requires the Mertens theorem with a remainder that is uniform in the upper limit.
- Control of the Chebyshev sum $\sum_{p \le N^{\theta_0}} \frac{\log p}{p}$ with an error term $O(1)$ that does not depend on θ_0 when θ_0 is bounded away from 0.
- A rigorous expansion of the ratio $L(N, \theta_0)/T(N)$ to order $1/\log\log N$, keeping track of all lower-order terms that might contribute to the constant C.

These are non-trivial but appear feasible with current sieve and analytic methods; the present empirical work provides a clear target for such an investigation.

7 Refined Prefactor and Empirical Validation

Table 3 compares the naive prefactor $\log(1/\theta_0)$, the heuristic $f(\theta_0)$, and the empirical $b_{\text{eff}} = \delta \cdot \log\log N$ averaged over $N \in [10^4, 10^8]$.

The heuristic reduces the maximum relative error from 4.3× (naive) to under 1.21× across all eight thresholds. The remaining monotone trend in $b_{\text{eff}}/f(\theta_0)$ (from 0.90 to 1.20) suggests further subleading structure, partly due to the discrete activation thresholds which are most influential at small θ_0.

Table 3: Empirical b_{eff} (mean of $\delta \cdot \log\log N$ over $N \in [10^4, 10^8]$) vs. the naive prefactor $\log(1/\theta_0)$ and the heuristic $f(\theta_0) = \log(1/\theta_0) - \theta_0 - \log 2 + \frac{1}{2}$ (Conjecture 1). Standard deviations of b_{eff} are also shown.

θ_0	$\log(1/\theta_0)$	$f(\theta_0)$	b_{eff}	std dev	ratio (naive)	ratio (heuristic)
0.10	2.303	2.009	1.819	0.096	0.790	0.905
0.15	1.897	1.554	1.465	0.071	0.772	0.943
0.20	1.609	1.216	1.205	0.050	0.749	0.991
0.25	1.386	0.943	0.976	0.031	0.704	1.034
0.30	1.204	0.711	0.763	0.026	0.634	1.074
0.35	1.050	0.507	0.562	0.023	0.535	1.108
0.40	0.916	0.323	0.373	0.017	0.407	1.154
0.45	0.799	0.155	0.187	0.010	0.234	1.202

Remark 3. $f(\theta_0)$ approaches zero near $\theta_0 \approx 0.47$; the $1.21\times$ bound applies only for $\theta_0 \in [0.10, 0.45]$ and should not be extrapolated.

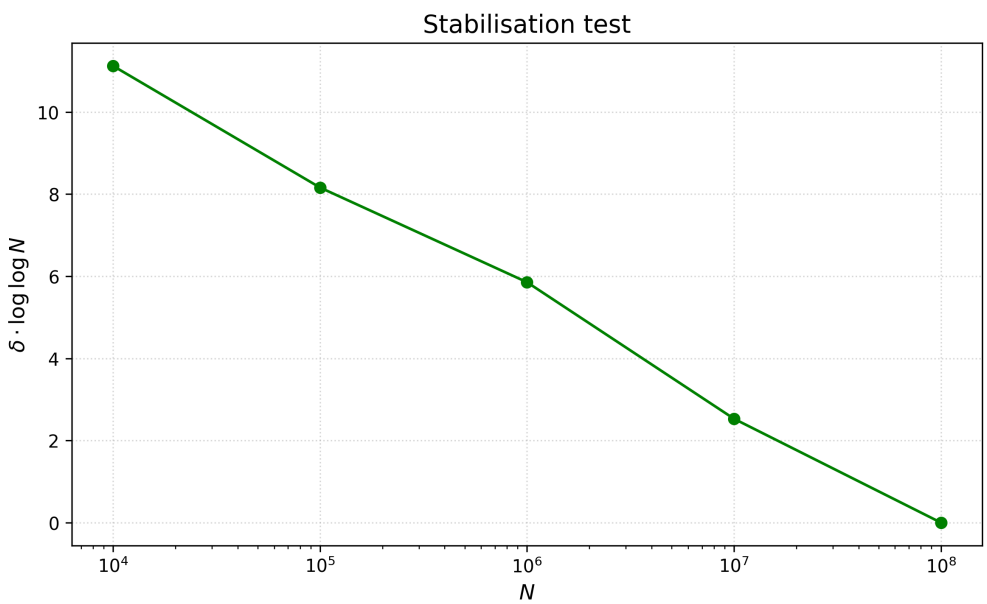

Figure 3: $b_{\text{eff}}(N, \theta_0) = \delta \cdot \log\log N$ vs. N for $\theta_0 \in \{0.20, 0.25, 0.30, 0.35\}$. Dashed lines show the heuristic value $f(\theta_0)$. Near-flat curves support Conjecture 1 within this range.

7.1 Possible exact constant and generalisation to k-almost primes

The constant $C = \frac{1}{2}$ has a natural interpretation: it arises from the Chebyshev sum $\sum_{p \le \sqrt{N}} (\log p)/p = \frac{1}{2}\log N + O(1)$, i.e., the midpoint of the $\log p$ distribution when the cutoff is $\sqrt{N}$. For k-almost primes $n = p_1 p_2 \cdots p_k$ with $p_1 \le p_2 \le \cdots \le p_k$, the natural generalisation of θ is

$$\theta_i = \frac{\log p_i}{\log n}, \qquad \sum_{i=1}^{k} \theta_i = 1.$$

The deficit for the smallest factor $\theta_1 \leq \theta_0$ would involve an iterated Mertens expansion with cutoffs $N^{\theta_0}, N^{\theta_0/(1-\theta_0)}$, etc., and the constant term at each stage generalises the $\frac{1}{2}$ to a sum over Chebyshev midpoints. This suggests a recursive structure that could be tested computationally for $k = 3$ (the next simplest case after semiprimes).

8 Geometric and Spectral Tests

We tested several candidate embeddings to see if the factor-asymmetry structure carried by θ could be recovered from alternative representations. Table 4 summarises the methods and outcomes. In every case, either the signal was dominated by trivial log-density effects (more semiprimes at larger N produce stronger signals) or by discrete artifacts from very small primes ($p \in \{2, 3, 5, \ldots, 31\}$). No stable localization by θ was observed.

Table 4: Summary of embeddings tested and observed outcomes. Full details are given in Appendix B.

Embedding	**Outcome**
Additive phase wrapping $\phi_t(n) = t \log n \bmod 2\pi$	No stable localization; patterns explained by log-density or small primes
Multiplicative phase maps n^{it}	Similar to additive case; no clustering
Spectral fingerprint $C(t) = \left\lvert \sum e^{it \log n} \right\rvert / N$ over semiprime subsequence	No separation by θ
Factor-ratio spectrum $C_{\text{ratio}}(t) = \left\lvert \sum e^{it \log(p/q)} \right\rvert / N$	Structure collapses to small-prime artifacts

This failure is not accidental. Any function of n alone cannot distinguish between the two factorizations of a semiprime; it sees only the product. The coordinate θ, by contrast, depends explicitly on the factorization through $\log p / \log n$. A separation principle is suggested: factor asymmetry is not recoverable from any embedding that is multiplicative or additive in log-space and depends only on n. Proving such a statement would require a formal characterization of the class of functions of n that are invariant under the symmetry $p \leftrightarrow q$.

Thus θ emerges as the *canonical* coordinate for the factor asymmetry problem, analogous to the use of projective coordinates in geometry or reduced fractions in Farey sequences.

9 Results and Multi-Scale Synthesis

9.1 Summary of empirical findings

For fixed $\theta_0 \in (0, \frac{1}{2})$, the following is consistent with the data:

1. $P(N, \theta_0)$ increases monotonically across the sampled powers of ten for all four tested thresholds.

2. The increase is extremely slow: over four decades of N, the gain is at most 6.8% (at $\theta_0 = 0.20$), compared with the $\sim 50\%$ gap that remains at $N = 10^8$.

3. Within the observed range, the deficit $\delta(N, \theta_0) = 1 - P(N, \theta_0)$ decreases at a rate well approximated by a constant per unit $\log N$, with rate $b(\theta_0)$ as tabulated in Table 2.

4. Discrete activation thresholds at $N = p^{1/\theta_0}$ produce step-like upward jumps in $P(N, \theta_0)$, most visible at small θ_0.

9.2 Multi-scale interpretation

The observed behavior can be understood as the interaction of three distinct layers:

(1) **Smooth analytic layer (Mertens scale)**: The envelope $\sim 1/\log\log N$ captures the dominant trend from the accumulation of small prime contributions. This layer is continuous and arises from integral approximations to prime sums.

(2) **Discrete prime layer (activation scale)**: Each prime p activates at $N = p^{1/\theta_0}$ and contributes a finite jump. The superposition of these jumps, together with their slowly varying tails, produces the observed step structure. The spacing between jumps encodes prime gap statistics. This can be viewed as a *marked renewal process* in the variable $t = \log\log N$, where the inter-arrival times are roughly $(p_{k+1} - p_k)p^{1/\theta_0 - 1}$ and the marks Δ_p have size $\sim 1/((1-\theta_0)p\log\log N)$. For large N the unresolved tail of activations contributes a negligible drift, explaining why $b_{\text{eff}}/f(\theta_0)$ stabilises in Figure 3.

(3) **Geometric projection layer**: The coordinate $\theta = \log p/\log n$ projects the pair $(\log p, \log q)$ onto the unit interval, separating factor imbalance from overall magnitude. This projective structure explains why embeddings of n alone fail and suggests that θ is the natural coordinate for the problem.

The failure of smooth asymptotics to capture the full detail is not a deficiency of the analytic methods but rather a consequence of the multi-scale nature of the system. The convergence is a genuine superposition of continuous and discrete structures that interfere in the observed range.

9.3 Interpretation and connection to known theory

The extremely slow approach is qualitatively consistent with what analytic number theory predicts for ratio-constrained semiprimes. The density of semiprimes $n = pq \le N$ with $p \le N^u$ for fixed $u < \frac{1}{2}$ is governed by sums of the form $\sum_{p \le N^u} \pi(N/p)$. Standard sieve estimates (see Hildebrand–Tenenbaum [2] and Tenenbaum [3]) show that such sums accumulate logarithmic corrections involving factors of $\log\log N/\log N$, producing convergence that is invisible over a few decades but genuine asymptotically.

The local linear fit (3) captures the dominant behavior in the observed range $[10^4, 10^8]$, but it is a finite-range approximation: the true deficit decays to zero more slowly than any fixed linear function of $\log N$ would predict, and the functional form of $P_\infty(\theta_0)$ is not determined by our data. What we can say is that the data are consistent with the existence of a limiting value $P_\infty(\theta_0)$ (which the proportion cannot exceed 1), approached from below. Whether this limit equals 1 for all $\theta_0 < 1/2$ is a plausible conjecture (see abstract and sieve heuristics), but not proved here.

9.4 The θ coordinate

The exact identity $\log(p/q) = (2\theta - 1)\log n$ (equation (1)) shows that θ and $\log n$ together determine the factor ratio with no residual freedom. The projective interpretation $\theta = \log p/(\log p + \log q)$

makes θ a natural coordinate: it separates the role of the factor imbalance (encoded in θ) from the role of the magnitude of n (encoded in $\log n$). The observed convergence is therefore a statement about how the distribution of θ across semiprimes $n \leq N$ evolves as N grows, a well-posed and nontrivial question even if its full resolution requires deeper sieve-theoretic tools.

10 Conclusion

Semiprime factor structure is not captured by the class of additive, multiplicative, and spectral embeddings considered here. The logarithmic coordinate $\theta = \log p / \log n$ provides the most stable description of factor asymmetry, and its projective nature explains why simpler embeddings fail.

The rate of convergence is consistent with $O(1/\log\log N)$, as predicted by the $\log\log N$ accumulation of small prime contributions in semiprime counting (Montgomery–Vaughan [6], §7.4). The local linear-in-$\log N$ approximation of Section 5.3 is a valid finite-range description of this slower decay: over $N \in [10^4, 10^8]$, $\log\log N$ varies by only 0.69, so the two functional forms are numerically indistinguishable in this window.

The novel contributions are: (1) empirical quantification of the convergence rate $b(\theta_0)$ across four imbalance thresholds, suggesting it decreases smoothly with θ_0; (2) the discrete activation threshold mechanism (Section 4), interpreted here as a prime-driven renewal process with a decomposition into smooth envelope plus explicit jump sum; (3) the geometric and spectral null results, consistent with a separation principle: factor asymmetry is not recoverable from any function of n alone that is multiplicative or additive in log-space; (4) a second-order heuristic for the prefactor $f(\theta_0) = \log(1/\theta_0) - \theta_0 - \log 2 + \frac{1}{2}$ (Conjecture 1, Section 6), where the constant $\frac{1}{2}$ encodes the balanced split point $p = q$; and (5) a multi-scale synthesis showing that the observed behavior arises from the interaction of smooth analytic, discrete prime, and geometric projection layers.

Future directions. Several natural extensions remain for future work:

- Push the exact enumeration to $N = 10^9$ or 10^{10} using segmented sieves and optimized counting; this would test whether the mild monotone trend in $b_{\text{eff}}/f(\theta_0)$ continues or stabilises.
- Perform a denser sampling of θ_0 values (e.g., $\theta_0 = 0.10, 0.12, \ldots, 0.45$) to better resolve the functional form of the convergence rate and to validate the heuristic prefactor across a finer grid.
- Quantify the contribution of the largest activation steps to the residual drift in $b_{\text{eff}}/f(\theta_0)$; this would help separate discrete prime effects from smooth subleading asymptotics.
- Formalize the renewal decomposition $X(N) = X_{\text{smooth}}(N) + \sum \Delta_p(N)$ and establish rigorous error bounds on the remainder.
- Study the boundary layer $\theta \sim 1/\log\log N$ with scaling variable $u = \theta \log\log N$. This regime may exhibit a nontrivial limiting distribution analogous to the Dickman or Buchstab functions, representing a new "semiprime imbalance scaling function."
- Prove the separation principle: no function of n alone that is multiplicative or additive in $\log n$ can recover θ for a positive proportion of semiprimes.
- Compare the observed distribution more systematically against the full asymptotic formulas from the literature on numbers with a smallest prime factor $\leq x^\alpha$ (Hildebrand–Tenenbaum, Tenenbaum).

- Quantify the contribution of twin primes and other prime-pair constellations to the discrete jumps at activation thresholds, linking to Cramér-type models of prime gaps.
- Extend the analysis to k-almost primes ($k \geq 3$), where the projective coordinates θ_i satisfy $\sum \theta_i = 1$ and the deficit for the smallest factor involves iterated Mertens expansions.

Proving Conjecture 1, extending computations to $N = 10^{10}$, and determining $P_\infty(\theta_0)$ rigorously remain open problems accessible to sieve-theoretic methods. The negative results on geometric and spectral embeddings suggest that θ is not merely a convenient coordinate but is in some sense the correct one for this problem. The multi-scale decomposition proposed here offers a roadmap for turning empirical observations into a structural theory.

All data and code are available at https://github.com/onojk/primehelix.

References

[1] K. Dickman, "On the frequency of numbers containing prime factors of a certain relative magnitude," *Arkiv för Matematik, Astronomi och Fysik*, vol. 22A, no. 10, 1930.

[2] A. Hildebrand and G. Tenenbaum, "Integers without large prime factors," *Journal de Théorie des Nombres de Bordeaux*, vol. 5, pp. 411–484, 1993.

[3] G. Tenenbaum, *Introduction to Analytic and Probabilistic Number Theory*, Cambridge University Press, 1995.

[4] P. Erdős and C. Pomerance, "On the distribution of smooth numbers," in *Number Theory, Carbondale 1979*, Lecture Notes in Math. 751, Springer, Berlin, 1979, pp. 173–183.

[5] H. Halberstam and H.-E. Richert, *Sieve Methods*, Academic Press, London, 1974.

[6] H. L. Montgomery and R. C. Vaughan, *Multiplicative Number Theory I: Classical Theory*, Cambridge Studies in Advanced Mathematics 97, Cambridge University Press, 2007.

A Heuristic Derivation Details

We sketch the heuristic expansions for $L(N, \theta_0) = \sum_{p \leq N^{\theta_0}} (\pi(N/p) - \pi(p-1))$ and $T(N) = \sum_{p \leq \sqrt{N}} (\pi(N/p) - \pi(p-1))$. Using the prime number theorem in the form $\pi(x) = \frac{x}{\log x} + \frac{x}{(\log x)^2} + O\left(\frac{x}{(\log x)^3}\right)$, and approximating sums over primes by integrals via Mertens' theorem, we obtain:

$$\sum_{p \leq x} \frac{1}{p} = \log\log x + M + o(1), \qquad \sum_{p \leq x} \frac{1}{p \log p} = \frac{1}{\log x} + o\left(\frac{1}{\log x}\right).$$

For $L(N, \theta_0)$, write

$$L(N, \theta_0) = \sum_{p \leq N^{\theta_0}} \pi(N/p) - \sum_{p \leq N^{\theta_0}} \pi(p-1).$$

The second sum is of lower order because $\pi(p-1) \sim p/\log p$, and its contribution is $O(N^{\theta_0}/\log N)$. For the first sum, substitute the asymptotic for π:

$$\sum_{p \leq N^{\theta_0}} \frac{N/p}{\log(N/p)} + \frac{N/p}{(\log(N/p))^2}.$$

Factor $N/\log N$:

$$\frac{N}{\log N}\sum_{p\le N^{\theta_0}}\frac{1}{p}\cdot\frac{\log N}{\log(N/p)}+\frac{N}{(\log N)^2}\sum_{p\le N^{\theta_0}}\frac{1}{p}\cdot\frac{(\log N)^2}{(\log(N/p))^2}.$$

For $p\le N^{\theta_0}$ with $\theta_0<1$, we have $\log(N/p)=\log N-\log p$. Writing $\log(N/p)=\log N(1-\frac{\log p}{\log N})$, and expanding the geometric series, we get $\frac{\log N}{\log(N/p)}=1+\frac{\log p}{\log N}+O\Big(\frac{(\log p)^2}{(\log N)^2}\Big)$. Thus

$$\sum_{p\le N^{\theta_0}}\frac{1}{p}\cdot\frac{\log N}{\log(N/p)}=\sum_{p\le N^{\theta_0}}\frac{1}{p}+\frac{1}{\log N}\sum_{p\le N^{\theta_0}}\frac{\log p}{p}+\cdots.$$

The first sum gives $\log\log N+\log\theta_0+M+o(1)$. The second sum is $\sum_{p\le N^{\theta_0}}\frac{\log p}{p}=\log N^{\theta_0}+O(1)=\theta_0\log N+O(1)$ by Chebyshev's estimate $\sum_{p\le x}\frac{\log p}{p}=\log x+O(1)$. Hence the $\frac{1}{\log N}$ factor turns this into $\theta_0+O(1/\log N)$. Collecting terms yields

$$L(N,\theta_0)\approx\frac{N}{\log N}\big(\log\log N+\log\theta_0+M+\theta_0\big).$$

For $T(N)$, the cutoff is $\sqrt{N}$. The same expansion with $\theta_0=1/2$ gives

$$\sum_{p\le\sqrt{N}}\frac{1}{p}=\log\log\sqrt{N}+M+o(1)=\log\log N-\log 2+M+o(1).$$

and

$$\frac{1}{\log N}\sum_{p\le\sqrt{N}}\frac{\log p}{p}=\frac{1}{\log N}\big(\log\sqrt{N}+O(1)\big)=\tfrac{1}{2}+O(1/\log N).$$

Thus

$$T(N)\approx\frac{N}{\log N}\big(\log\log N-\log 2+M+\tfrac{1}{2}\big).$$

These heuristic expansions ignore error terms from the prime number theorem, from the truncation of the series expansion, and from the moving cutoff. The empirical agreement suggests that the main constant terms are captured correctly.

B Details of Geometric and Spectral Tests

Additive phase wrapping. For a fixed parameter t, define $\phi_t(n)=t\log n \bmod 2\pi$. For a set of semiprimes n, the resulting points on the unit circle were examined for clustering by θ. No stable localization was found; the empirical distribution of phases was essentially uniform except for artifacts from the smallest semiprimes, which are too few to be meaningful.

Multiplicative phase maps (Fourier on the multiplicative group). The sequence $n^{it}=e^{it\log n}$ was summed over semiprime indices. The resulting sums were dominated by the gross count of semiprimes, and no dependence on θ could be extracted.

Spectral fingerprint. We computed the normalized power spectrum $C(t)=\big|\sum_{\text{semiprime } n\le N}e^{it\log n}\big|/\pi_2(N)$ for a range of t. The Fourier transform of the semiprime indicator does not separate by θ because the transform mixes all numbers; the only visible features are at frequencies corresponding to small primes.

Factor-ratio spectrum. The same construction was applied to the factor ratio $\log(p/q)$: $C_{\text{ratio}}(t) = \left|\sum e^{it\log(p/q)}\right|/\pi_2(N)$. This collapses to a few dominant frequencies originating from the smallest primes (e.g., $p = 2, q = 3$ gives $\log(2/3)$), but does not reveal any continuous dependence on θ.

A more detailed account, including plots, is given in the online repository.

C Boundary-Layer Scaling (Speculative)

This direction is speculative and is included for completeness; it makes no empirical contact with the data in the main text.

Introduce the rescaled variable $u = \theta \log\log N$. For θ of order $1/\log\log N$, a uniform expansion is conjectured (details in the online repository). This boundary layer corresponds to the regime where the smaller factor p is extremely small compared to n, and the deficit structure transitions from the $f(\theta_0)/\log\log N$ form to a different limiting distribution. The discrete activation thresholds become particularly pronounced in this region. A rigorous treatment would require a careful Mellin-transform analysis of the counting sum $L(N, \theta_0)$ in the boundary regime, which we leave for future work.